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Statement of Purpose: Self-Discovery and Long-Term Well-Being Empowered by Human-AI Interaction

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The Root. In many traditional East Asian homes, conversations about the inner life—feelings, identity, and mental health—are treated like a "Forbidden City," avoided in favor of the familiar myth that if you get into a good school and find a good job, everything else will simply sort itself out. I followed that script for a time. After completing my undergraduate degree at Columbia, I began my career in investment research on Wall Street, where the endless hamster wheel of advancement often blurred the questions of meaning, direction, and identity that many quietly share. In the margins of my week, I began my own experiments—reading across disciplines, prototyping small engineering projects, and informally tracking and intervening on my own behavior—and became fascinated by **how different modalities can reflect distinct facets of the self**. As Epictetus argued, "No man is free who is not master of himself." Today, at a time when AI systems can either amplify a "digital panopticon" or help us turn inward with greater clarity and connection, I am committed to the latter: at Columbia, I hope to explore **Human–AI Interaction for self-discovery through behavioral interventions, physiological sensing, and creative expression**.

AI Agents for Self-Exploration and Introspection: Self-clarity is one of the most reliable psychological correlates of life satisfaction, emotional stability, and effective self-regulation, yet access to deep introspective experiences remains inaccessible and unevenly distributed. I am excited about **AI agents that serve as companions for self-exploration**, making introspective practices—such as mindfulness, self-reflection, and critical thinking—more accessible as forms of everyday psychological support. Leading the *MindfulAgents* project [1] has thoroughly prepared me for this line of work in terms of **AI agent prototyping, intervention evaluation, and multi-party collaboration**. In partnership with Equa Health, I designed and shipped a real-world multi-agent system to improve long-term engagement with mindfulness via a consumer-facing app. Designing and leading a mixed-method evaluation—including a lab study, a four-week in-the-wild deployment with 60+ participants, and qualitative interviews—trained me to structure rigorous studies and synthesize quantitative and qualitative evidence. Extending beyond this project, I am particularly interested in (1) **socially intelligent agents** that can sustain engagement with introspective practices by adapting to user affect, goals, and context; (2) AI-assisted, value-grounded daily planning systems that **connect daily decisions with a person's longer-term values**; and (3) **AI "twin" agents** that help users see themselves from multiple perspectives. The effectiveness of such agents, however, is contingent upon a rich, continuous understanding of a user's real-world behavior.

Optimizing Behavioral Patterns through the Quantified Self: "We are what we repeatedly do. Excellence, then, is not an act, but a habit." Everyday behavior is where self-identity and core values either take root or erode, and I am interested in designing **quantified-self systems with ubiquitous sensing that help people see and reshape behavioral patterns**. My prior work has equipped me to operate at this intersection of behavior and physiological data: during a summer research intern with Prof. Sam Gershman, I applied reinforcement-learning theory in a Bayesian framework across five large-scale multimodal health datasets to test hypotheses about anhedonia and Parkinson's disease. At the LIINC Lab, I contributed to a physiological foundation model effort: EEG/wearable modality preprocessing, pretraining and fine-tuning pipelines, and Optuna-based evaluation, which gave me deep familiarity with ML for physiological signals and a solid grounding in signal processing and passive sensing [2]. Looking ahead, I plan to explore: (1)

multimodal data streams and reasoning in daily life; (2) how integrating **daily sensing along underexplored dimensions**—such as mindfulness practice, social energy, and focus—can support deeper insight; and (3) how interventions for **behavioral addiction and habit formation can leverage AI and ubiquitous sensing**. While quantified data provides a powerful starting point, its potential for self-connection might amplify when it is translated into creative forms.

AI-assisted Creative Manifestation of Self: Creative forms, our aesthetics, homes, objects, and visualizations, are powerful mirrors of identity, and for many people it is easier to approach themselves through images and spaces than through direct "talk therapy." I am interested in **AI systems that use creative media—including generative art, interior designs, and XR environments—as alternative interfaces for self-exploration**. Under Prof. Iigaya, I studied how aesthetic stimuli evoke emotion and relate to psychiatric traits, sharpening my skills in affective computing and representation learning. Through my work on *SwEYEpinch* [3] with Prof. Steve Feiner, I gained experience in VR/AR system design and experimentation, designing a gaze-based XR text-entry system and leading four multi-visit user studies. Building on this foundation, I am drawn to exploring: (1) **AI-assisted interior design systems** inspired by danshari—mindful, value-based decluttering that help people configure spaces aligned with who they are becoming; (2) **transforming behavioral traces**—for example, workout or sleep trajectories—**into generative visualizations** that make effort and fluctuation tangible, serving as reflective tools and motivational anchors; and (3) **XR-based spaces for self-containment**, allowing people to inhabit visual representations of their inner and outer transformations.

Vision Beyond: After the Ph.D., I aspire to become a principal investigator leading a **cross-disciplinary group at the intersection of cognitive science, human–computer interaction, and ubiquitous sensing**. I hope to measure impact not only by publications, but also by deployable systems in real use, developed in partnership with clinicians, health experts, and community stakeholders. I am also committed to sharing knowledge, having benefited enormously from mentors who opened both technical and personal pathways for me, and I hope to pay that forward through teaching and mentorship.

References

[1] **Wu, Mengyuan**, Fan, Y.*, Jiang, Z.*, Feng, R., Dharmavaram, S., Fallon, S., Polowitz, M., Islam, B., Benson, L., Tung, I., Creswell, J. D., Xu, X. MindfulAgents: Personalizing Mindfulness Meditation via an Expert-Aligned Multi-Agent System. ACM CHI 2026 (Honorable Mention Award).

[2] Li, Z., Rong, K., **Wu, Mengyuan**, Yamauchi, A., Chen, H., Peng, Y., Tao, T., Sajda, P. *Physiological Foundation Models for Robotic Attention Reorientation.* Manuscript In Preparation.

[3] Li, Z.*, He, X.*, **Wu, Mengyuan**, Tong, Z., Wei, H., Yang, B., Sajda, P., Feiner, S. SwEYEpinch: Exploring Intuitive, Efficient Text Entry for Extended Reality via Eye and Hand Tracking. ACM CHI 2026.